GREENFIELD PUBLIC SCHOOL, CHENNAI ### Class 11 Physics — Unit Test 3 — Portions, Blueprint and Draft Question Paper (Fictional school for training purposes)

Prepared by: Mrs. Kavitha R (PGT Physics) — **DRAFT, not yet checked against the blueprint**
Test date: Thursday, 3 December 2026 · 8:45 to 9:45 am · 25 marks

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Part 1. UT3 portions (NCERT Class 11 Physics)

- Chapter 5: Work, Energy and Power
- Chapter 6: System of Particles and Rotational Motion
- Chapter 7: Gravitation
- Chapter 8: Mechanical Properties of Solids

Part 2. Approved department blueprint (final, 20 October 2026)

Paper pattern: Section A — 5 MCQs × 1 mark; Section B — 4 short answers × 2 marks; Section C — 4 answers × 3 marks. Total 25 marks. At least 5 marks (20%) must be competency-based (case-based / application).

Chapter	MCQ (1 mark)	Short answer (2 marks)	3-mark answer	Total marks
5 Work, Energy and Power	1	0	2	**7**
6 System of Particles and Rotational Motion	2	2	0	**6**
7 Gravitation	0	2	1	**7**
8 Mechanical Properties of Solids	2	0	1	**5**
Total	**5**	**4**	**4**	**25**

Part 3. Draft question paper

Section A — Multiple choice (1 mark each)

1. The SI unit of power is: (a) joule (b) watt (c) newton (d) pascal
2. The moment of inertia of a body depends on: (a) its mass only (b) the distribution of its mass about the axis (c) its angular velocity (d) the torque on it
3. The SI unit of torque is: (a) N m (b) N/m (c) J/s (d) kg m/s
4. The acceleration due to gravity at the centre of the Earth is: (a) 9.8 m/s^2 (b) infinite (c) zero (d) 4.9 m/s^2
5. The escape velocity from the Earth's surface is about: (a) 7.9 km/s (b) 11.2 km/s (c) 3 km/s (d) 30 km/s

Section B — Short answer (2 marks each)

1. State the principle of conservation of angular momentum and give one example.
2. Why is it easier to open a door by pushing at the handle than near the hinges?
3. State Kepler's law of areas.
4. State the work-energy theorem.

Section C — Answer in 3 to 4 sentences or steps (3 marks each)

1. *(Case-based)* A figure skater spinning with her arms stretched out pulls her arms close to her body. What happens to her angular speed? Explain why.
2. Derive an expression for the acceleration due to gravity at a small height h above the Earth's surface (h much smaller than R).
3. A body of mass 2 kg moving at 10 m/s is brought to rest by friction. How much work is done by friction?
4. *(Case-based)* Communication satellites used for DTH television appear fixed in the sky. Explain why such a satellite must orbit above the equator with a period of 24 hours, and give one use of it.

Answer key and marking scheme: to be attached before submission to the Exam Cell.